

DOMAIN-SPECIFIC LEGAL LLM

SecondLayer Legal AI Model as a Service

We trained a 14B-parameter LLM on 115M Ukrainian court decisions. Now we sell inference via API -- so any legal tech product can have expert-level legal reasoning without training their own model.

115M court decisions cleaned & tokenized

8xH100 GPU + MLflow pipeline

First in Ukraine a 500B model costs \$10M+ to replicate

legal.org.ua | [AWS Activate](#) | [Google for Startups](#) | [NVIDIA Inception](#)

The Problem

01

Generic LLM + RAG Hits a Ceiling

Plugging court docs into ChatGPT looks like it works -- until it doesn't. RAG retrieves documents, but the model still confuses procedural stages, hallucinates citations, and misreads Ukrainian legal reasoning. **The model was never trained on this domain** -- it pattern-matches on English common law, not Ukrainian civil code. Only domain-trained weights fix this.

02

Training Is a 12-18 Month Barrier

Any legal tech startup that wants reliable AI needs domain-specific training -- data pipelines, GPU infrastructure, ML expertise. Most can't afford it and shouldn't have to.

03

No Legal LLM for CEE Exists

Harvey (US), Luminance (UK) -- closed, English-only, \$100K+ contracts. Zero options for Ukrainian, Polish, Romanian, or other CEE jurisdictions.

04

\$15B Legal AI Market, No Infrastructure Layer

Everyone builds apps. Nobody provides the foundational model. Legal tech needs its own inference layer -- like Stripe is to payments, we are to legal reasoning.

The **Solution**: Legal LLM Inference API

One API call. Expert-level Ukrainian legal reasoning. No training required by the customer.

Completion API

OpenAI-compatible endpoint. Drop-in replacement for legal tasks. Chat, analysis, extraction, summarization.

RAG-Enhanced Retrieval

Model + 2M vector embeddings + 5M full-text court decisions. Grounded answers with citations.

Specialized Endpoints

Case outcome prediction, clause risk scoring, contract review, precedent matching.

- **legal.org.ua** -- our own product, built on this model. 118 AI tools in production. Proves the API works before customers plug in.

HOW IT WORKS

POST /v1/chat/completions

```
{
  "model": "legalex-14b",
  "messages": [{
    "role": "user",
    "content": "Analyze liability
under Art. 625 CCU..."
  }]
}
```

Latency

<2s first token

Context

32K tokens

Format

OpenAI-compatible

The Model

Not a roadmap -- **active training**. CPT on Qwen2.5-14B running on 8xH100 right now. Every experiment tracked in MLflow.

● TRAINING NOW

CPT Qwen2.5-14B

Continual pre-training on 115M court decisions. 8xH100 via NVIDIA Innovation Lab. ~48s/step.

● COMPLETE

RAG + Embeddings

2M+ domain vectors indexed. Semantic retrieval over 5M full-text court decisions. Production-ready.

● COMPLETE

Temporal Drift Study

24-job matrix. 12 models, 9 families. Proved GPT-4o degrades on post-training legal texts.

● NEXT

Scale to 32B

Full EDRSR corpus. Target: surpass GPT-4o on Ukrainian legal benchmarks.

TRAINING DATA

115M

Court decisions (full EDRSR)

200K+

Laws and regulations

2M+

Vector embeddings indexed

INFRASTRUCTURE

8xH100 (NVIDIA Innovation Lab)

MLflow experiment tracking

SageMaker GPU pipelines

HuggingFace Hub (8 datasets)

Why **Us**, Why Now

DATA MOAT

115M decisions

Cleaned, deduplicated, tokenized

Custom pipeline

Domain tokenizer, legal-specific cleaning

Data is public -- but our pipeline took 18 months to build

RESEARCH MOAT

9 papers

4 on arXiv, peer-reviewed

PhD in progress

NAS Ukraine, Glushkov Institute

Published proof that our approach works

TIMING MOAT

First mover

Zero CEE legal LLMs exist today

12-18 months

Head start over any new entrant

Model compounds -- each jurisdiction adds network value

legal.org.ua -- our own SaaS, built on this model

118 AI tools live. Proves product-market fit and generates first revenue while inference API scales.

Live Product

118 AI Tools

50+ MLflow Runs

Business Model: Inference Revenue



Market Opportunity

TAM **\$15B**

SAM **\$200M**

SOM **\$5M**

TAM -- Legal AI Infrastructure

Global legal AI spending is **\$15B by 2027** (Grand View Research). Model inference is the infrastructure layer -- every legal AI product is a potential customer.

SAM -- CEE + Ukrainian Legal Tech

Ukraine (60K+ lawyers, 1000+ law firms), Poland, Romania, Czech Republic. **\$200M** combined legal tech + compliance AI spend.

SOM -- Year 2 Target

\$5M ARR from 50-100 API customers averaging \$4-8K/mo. Plus enterprise deployments at \$2-5K/mo committed throughput.

Expansion Path

Each new jurisdiction = new model checkpoint + new customer base. **Poland next** (similar legal system, EU accession corpus).

Traction & Roadmap

14B

MODEL TRAINING ON 8XH100

9

RESEARCH PAPERS (4 ARXIV)

118

AI TOOLS IN PRODUCTION

3

STARTUP PROGRAMS (AWS, GCP,
NVIDIA)

NOW

14B model training completes. Benchmark against GPT-4o on UA legal tasks. API alpha with 5-10 design partners.

MONTHS 1-3

API public launch. Scale to 32B. NVIDIA Inception marketplace listing. First 20 paying API customers. \$5-15K MRR.

MONTHS 4-9

\$50K+ MRR. Enterprise contracts. Polish legal model training begins. Seed round. 50+ API customers.

NVIDIA Inception Member

\$100K AWS Credits

\$350K GCP Credits (pending)

Infra Cost = \$0 for 18+ months

Team & The Ask

VO

Volodymyr Ovcharov

CEO & AI Researcher

15+ years data platforms. NAS
Ukraine PhD. 9 papers. Trained
models on H100. Built the full
stack solo.

IK

Igor Kyrychenko

Co-Founder & CLO

PhD in Law. Domain expert.
Annotation quality, legal
modeling, compliance strategy,
customer relationships.

WHY \$50K IS ENOUGH

\$100K AWS credits -- infra free 18+ mo

NVIDIA Inception -- H100 training at \$0

\$350K GCP (pending) -- scale runway

Solo technical founder -- \$0 overhead

\$50,000 for 5% equity

\$1M pre-money valuation

60% Founder runway (6 months full-time)

25% Go-to-market & first API customers

15% Legal & operations

CONTACT

volodymyr@legal.org.ua

linkedin.com/in/volodymir-ovcharov

legal.org.ua

igor@legal.org.ua

+380 67 720 63 53